Multimodal Deep Learning Approach for Alzheimer's Disease Diagnosis

Cristina-Ioana CASAPU
Department of Computer Science and Information Technology
Faculty of Automation, Computers, Electrical Engineering and Electronics, Dunarea de Jos University of Galati
Galati, Romania
casapuioanacristina@yahoo.com

Simona MOLDOVANU
Department of Computer Science and Information Technology
Faculty of Automation, Computers, Electrical Engineering and Electronics, Dunarea de Jos University of Galati
Galati, Romania
simona.moldovanu@ugal.ro

2025 9th International Symposium on Electrical and Electronics Engineering (ISEEE) | 979-8-3315-7666-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISEEE67817.2025.11304814

***Abstract*— Early and accurate detection of Alzheimer's Disease (AD) is critical for timely intervention and improved patient outcomes. This paper introduces a multimodal deep learning framework for AD stage classification that combines structural magnetic resonance imaging (MRI) scans with clinical assessments, including the Mini-Mental State Examination (MMSE). The proposed system integrates three independently trained models: two convolutional neural networks (CNNs) with DenseNet121 backbones applied to distinct MRI datasets—processed as two-dimensional coronal and axial slices—and a third model based solely on clinical features. Features extracted from the CNN branches are processed through automated machine learning (AutoML) pipelines, and predictions are subsequently refined using four late-fusion strategies to enhance robustness. All models were trained using clinical data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), MRIs derived from the Open Access Series of Imaging Studies (OASIS), and additional public datasets. The best-performing configuration achieved classification accuracy above 90%, reaching up to 99% for moderate dementia, demonstrating the potential of explainable multimodal AI for reliable AD diagnosis.**

Keywords*— *Alzheimer's Disease, Transfer Learning, MRI, CNN, Multimodal Learning, AutoML, Late Fusion

I. Introduction

Alzheimer's Disease (AD) is the leading cause of dementia, affecting over 50 million individuals globally and ranking among the top contributors to age-related disability and dependency [1], [2]. Despite recent advances in the medical field, the diagnosis and staging of AD remain complex due to the subtle progression of cognitive decline. Early detection is critical for timely intervention, yet current diagnostic protocols often rely on isolated assessments—either clinical or structural—limiting predictive accuracy and scalability [3], [4].

Recent research emphasizes the value of multimodal machine learning approaches that integrate MRI imaging data with clinical scores such as the Mini-Mental State Examination (MMSE), for enhanced staging and personalized prediction [4], [5], [6]. However, key challenges persist, including class imbalance in public datasets, loss of spatial context in 2D CNNs, limited sample sizes for 3D models, and the lack of interpretability in AI-driven predictions [1], [2], [7]. Part of the data used in this study was provided by ADNI database (adni.loni.usc.edu), which has been instrumental in collecting and sharing standardized imaging and clinical datasets for Alzheimer's research. As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report (complete listing of ADNI investigators and their contributions can be found in the following paper https://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf).

This paper proposes an end-to-end multimodal system for Alzheimer's diagnosis and MMSE score prediction, based on the [8] thesis. The contribution consists of a multimodal system containing the followings: (1) a dual-branch CNN architecture trained on axial and coronal MRI slices (OASIS-1) and skull-stripped augmented data; (2) AutoML pipelines built from extracted CNN features; (3) four late fusion strategies—including Bayesian optimization—to combine predictions from the resulted models; and (4) a Gradient Boosted Trees model for MMSE regression using image and clinical inputs.

To enhance transparency, explainable AI features such as Grad-CAM and SHAP are integrated, for visual and feature-based interpretability. The system achieved a classification accuracy of 97.6% across four AD stages and an MMSE prediction R^2 of 81.2%, outperforming several recent state-of-the-art approaches thus these results highlight the potential of robust, explainable multimodal AI pipelines for clinical neurodegeneration diseases.

II. Current Studies

A. Deep Learning for Structural MRI Processing

Deep learning has become the dominant paradigm in classifying Alzheimer's Disease (AD) from brain imaging. The most of the existing approaches make use of 2D Convolutional Neural Networks (CNNs) trained on axial, coronal, or sagittal MRI slices [3]. Architectures such as DenseNet-169 and ResNet-50 have demonstrated solid performance, with DenseNet achieving up to 97.7% training accuracy in [3]. These models benefit from reduced computational cost and allow visual augmentations, but sacrifice 3D spatial coherence. To address this, 3D CNNs have been introduced for volumetric learning, capturing inter-slice dependencies across the brain [7], [9]. Studies using hippocampus-focused 3D CNNs or patch-based subvolume analysis report improved sensitivity in distinguishing Mild Cognitive Impairment (MCI) from AD [9]. Despite promising results, 3D models face challenges with standardization and limited labeled data.

B. Mini-Mental State Examination Score prediction

The Mini-Mental State Examination (MMSE) is a standardized 30-point cognitive assessment used to track AD progression. Recent works have explored MMSE prediction as a regression task using MRI and clinical features. Reference [4] showed that MMSE sub-scores—particularly temporal

979-8-3315-7666-0/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: Tsinghua University. Downloaded on January 12,2026 at 07:01:38 UTC from IEEE Xplore. Restrictions apply.

orientation and memory recall—are strong indicators of progression from MCI to AD. Other studies like [5] used speech and demographic features to estimate MMSE, achieving a mean absolute error of 4.97 using audio descriptors.

C. Multimodal Systems

A recent trend involves combining multiple data sources—MRI, PET, clinical scores—into unified multimodal systems. Reference [6] proposed an RVFL-based ensemble model fusing imaging with clinical/genetic features, reporting up to 95% accuracy on ADNI subsets. Transformer-based CNN hybrids and PET-MRI fusion models further enhance staging precision by capturing both metabolic and structural changes [1], [7]. These architectures demonstrate the power of cross-domain fusion in building clinically relevant Alzheimer's diagnostic tools.

III. Proposed Architecture

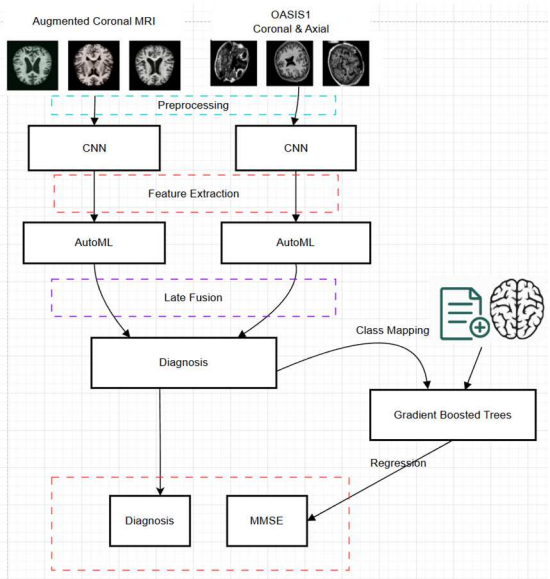

Fig. 1. System architecture [8]

The proposed system is a multimodal deep learning framework designed to classify Alzheimer's Disease stages and estimate MMSE scores by integrating structural MRI data with clinical features. As illustrated in Fig. 1, the architecture comprises two parallel CNN pipelines: one trained on raw axial and coronal slices from the OASIS-1 dataset, and another on skull-stripped and augmented coronal images. Each branch performs feature extraction independently, and the resulting embeddings are processed by an AutoML classifier to produce probabilistic predictions for different dementia stages.

To enhance classification accuracy and robustness, the system experiments with four late fusion strategies—including a Bayesian-optimized fusion layer—that aggregates predictions from both CNN and AutoML pipeline branches. The fused output yields a final diagnosis among four cognitive states. Additionally, a separate regression module based on Gradient Boosted Trees (version 3.0.5, accessed on 1st of June, 2025) is used to predict the patient's MMSE score using both imaging-derived and clinical features (specific MMSE questions related to performance on everyday tasks).

The architecture is implemented with the help of the following libraries: TensorFlow (version 2.19.0) and Keras (version 3.9.0) for CNN training, H2O AutoML for automated classification (version 3.46.0.7), and Scikit-learn (version 1.6.1) for regression and preprocessing, all the versions being available as of 1st of May, 2025. The entire pipeline is designed for modularity, reproducibility, and scalability across clinical datasets. Interpretability is ensured using Grad-CAM (version 1.0) for visual saliency mapping and SHAP (version 0.49.1) for regression insight, both accessed on 1st of June, 2025.

A. Datasets and Preprocessing

This study utilizes four complementary datasets to support both imaging-based classification and cognitive score prediction, the datasets sources can be found in [8]. For the image processing component, two datasets were constructed using OASIS-1 structural MRI data—one based on raw axial and coronal slices and another derived from skull-stripped coronal slices for focused attention on brain tissue (both datasets, sourced from the Kaggle platform, and not directly from OASIS-1, as mentioned in [8]). For MMSE score regression and 3-class Alzheimer's staging, the ADNI dataset, supplemented by an additional pre-augmented public dataset, as mentioned in [8], was utilized in order to improve the robustness and variability of training samples.

The first image dataset was derived from the publicly available OASIS-1 dataset, containing over 80.000 MRI slices [8]. These images were categorized into four classes: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented, following the labeling conventions in related work [3]. No skull stripping or augmentation was applied in this branch to preserve natural variance.

The second imaging dataset was constructed by applying skull stripping to the axial slices of OASIS-1 [8], removing non-brain tissue to enhance spatial focus. This dataset contains over 30.000 augmented coronal slices, distributed across the same four classes. Data augmentation includes horizontal flips, rotations, zoom, and contrast normalization. This branch served as the higher-performing input in the dual-branch architecture receiving higher fusion weight.

The main dataset used for regression was obtained from ADNI database (adni.loni.usc.edu). ADNI began in 2003 as a public–private partnership under the leadership of Michael W. Weiner, MD. Its initial aim was to investigate whether longitudinal measures—such as MRI, PET, biological markers, and cognitive assessments—could be combined to track the progression from mild cognitive impairment (MCI) to early Alzheimer's disease (AD). Over time, its objectives expanded to include the validation of biomarkers for clinical trials, broadening the diversity of study participants to enhance generalizability, and providing high-quality imaging and clinical data to the global scientific community. Further details are available at adni.loni.usc.edu.

The dataset includes MMSE scores, demographic variables (age, gender, education), and neurocognitive markers such as Frequently Asked Questions (FAQ) and Clinical Dementia Rating (CDR). A total of 10.000 samples were retained, with MMSE values ranging from 0 to 30. For classification purposes, these were grouped into three cognitive stages: CN – Cognitively Normal (28–30), MCI - Mild Cognitive Impairment (24–27), and AD – Alzheimer Disease (<24), based on established thresholds [3].

Authorized licensed use limited to: Tsinghua University. Downloaded on January 12,2026 at 07:01:38 UTC from IEEE Xplore. Restrictions apply.

An additional dataset containing over 3.000 rows of clinical data such as demographic variables, or health markers was used in order to enhance the previous dataset, as mentioned in the above dataset description. The dataset was cleaned and rebalanced, and mapped to corresponding MMSE scores.

It should be noted that MRI and clinical data were trained on separate datasets and combined only at the decision level within the multimodal framework.

B. CNN and AutoML training

The proposed classification framework for Alzheimer's disease employs a dual-branch convolutional neural network (CNN) architecture (Fig. 1), adapted from the approach in [10] originally designed for microorganism classification, and extended here to brain MRI data. One branch processes raw magnetic resonance images (axial and coronal views), while the second branch utilizes axially augmented MRI slices. In both CNN models, the first layer consist of the DenseNet121 network with frozen pre-trained weights owing to its proven capacity to capture salient structures in medical imaging through dense inter-layer connectivity, the rest of the CNN and AutoML architecture and hyperparameters being similar with the models in [10].

Training strategies were tailored to the data characteristics of each branch. As seen in Fig. 2, the augmented MRI branch was trained for 30 epochs with dropout regularization (rate = 0.5), early stopping, and learning-rate reduction on plateau, achieving stable validation accuracy of approximately 82–84% without significant overfitting, as seen in Fig. 2. The raw MRI branch followed a two-phase strategy: an initial three-epoch training on the full ~80,000-image set, followed by fine-tuning for seven epochs on a representative 30% subset. Although this approach accelerated convergence and promoted learning of fine-grained patterns, validation accuracy showed pronounced fluctuations (76–96%), attributed probably to the high variability inherent in augmented data.

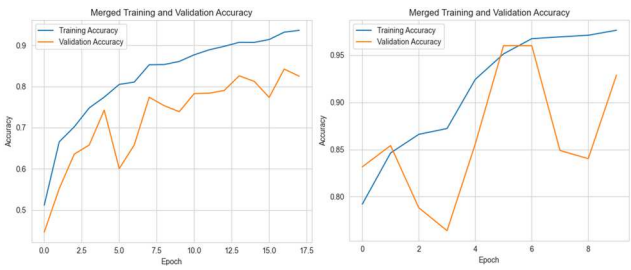

Fig. 2. First branch (last 18 epochs) and the raw MRI branch training and validation accuracy [8]

To address the instability observed in both CNN branches, feature vectors were extracted from the last BatchNormalization layer preceding the classification stage and subsequently processed through the H2O AutoML framework. This automated model-selection process evaluated multiple algorithms, similar with the configuration from [10], ultimately selecting Stacked Ensemble models for each branch. The augmented branch ensemble comprised exclusively deep learning models, while the raw branch ensemble integrated gradient boosting, distributed random forests, and deep learning components.

C. Late Fusion

The proposed system integrates predictions from the two above-mentioned complementary AutoML classifiers through a late fusion process. Each model outputs a probability vector where each element corresponds to one of the four diagnosis.

These vectors are combined element-wise according to four distinct weighting strategies, as follows (p_{fusion} being the final probability value). Firstly, in the global weight fusion approach, a single scalar weight (w) is applied uniformly to all classes as presented in (1), where p_1 is the probability from the first branch and p_2, from the other. Then, in the per-class weight strategy, a distinct weight is assigned to each class (2), where i is an index representing the class order ranging from 0 (Mild Dementia) to 3 (Very Mild Dementia). The third experimented fusion method is Bayesian optimization which refines these per-class weights using a Gaussian process to maximize accuracy, as in (2), but the weights are calculated with Bayesian optimization, while an alternative option optimizes the same weights according to the weighted F1-score (3), where y_{true} is the ground-truth label, and the argmax of p_{fusion} is the maximum fused probability.

$$p_{fusion} = w \cdot p_1 + (1 - w) \cdot p_2 \tag{1}$$

$$p_{fusion}[i] = w_i \cdot p_1[i] + (1 - w_i) \cdot p_2[\mathrm{i}] \tag{2}$$

$$argmax_{w_i} = F1(y_{true}, argmax(p_{fusion})) \tag{3}$$

After fusion, the resulting probabilities are mapped to the three ADNI categories: CN, MCI, and AD, by merging the Very Mild and Mild Dementia probabilities into the MCI class. The final diagnosis is determined as the class with the highest mapped probability.

D. MMSE Prediction

The MMSE cognitive score was estimated using a Gradient Boosted Trees (GBT) regression model implemented in XGBoost. Input features included FAQ-based functional scores, demographic variables (age, gender), and the diagnostic class predicted by the imaging classifier. The model was trained with an 80/20 split and tuned using a cross-validated search procedure that systematically identifies the best hyperparameters. This approach achieved strong predictive performance of cognitive decline that complements the classification component and enhances the overall diagnostic value of the system.

IV. Results

This section reports the experimental results obtained with the proposed system, focusing on both image-based classification and cognitive score prediction. The analysis begins with the impact of late fusion strategies, as long as the accuracy progress from the CNN-stage through each fusion method, including a detailed assessment of Bayesian optimization for weight selection. The section then presents the outcomes of MMSE regression, supported by error metrics and explainable AI analyses. Finally, the advantages of the multimodal framework are illustrated through representative case studies that combine imaging and clinical perspectives.

In the MMSE analysis section, several variables from the Functional Activities Questionnaire (FAQ) are used as predictors, including FAQFINAN (financial management), FAQFORM (form completion), FAQGAME (game and

Authorized licensed use limited to: Tsinghua University. Downloaded on January 12,2026 at 07:01:38 UTC from IEEE Xplore. Restrictions apply.

leisure activities), FAQBEVG (beverage preparation), FAQMEAL (meal preparation), FAQTV (television use), and FAQREM (memory tasks), together with demographic factors such as PTGENDER (gender).

A. Late Fusion

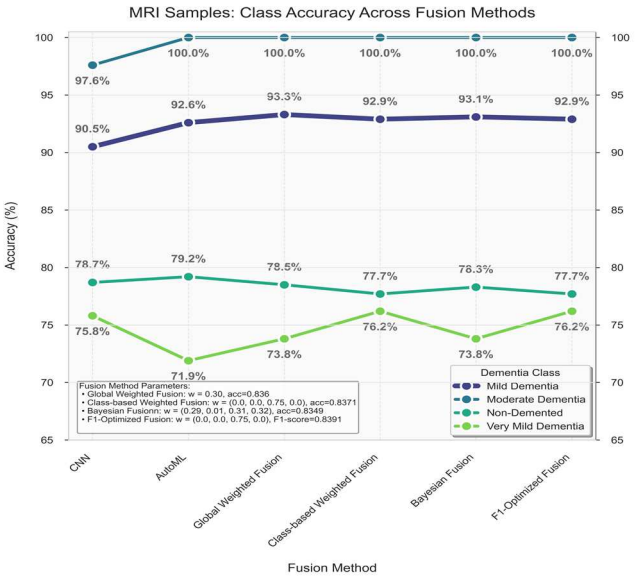

Fig. 3. Accuracy progress from the CNN stage through all the fusion stages for the raw MRI branch [8]

Late fusion provided consistent gains over the AutoML baselines by combining probability vectors from the two imaging branches. On raw MRI sections, as seen in Fig. 3, the most notable improvement was observed for Very Mild Dementia, where accuracy increased from 71.9% (AutoML) to 76.2% under class-based weighting. Mild Dementia also benefited, rising from 92.6% to 93.3% with global weighting, while Moderate Dementia remained saturated at 100% across all methods. A minor trade-off was identified for Non-Demented, which declined slightly from 79.2% to 77.7% across some fusion schemes, reflecting the well-known difficulty of separating this class from early-stage dementia in raw, non–skull-stripped images.

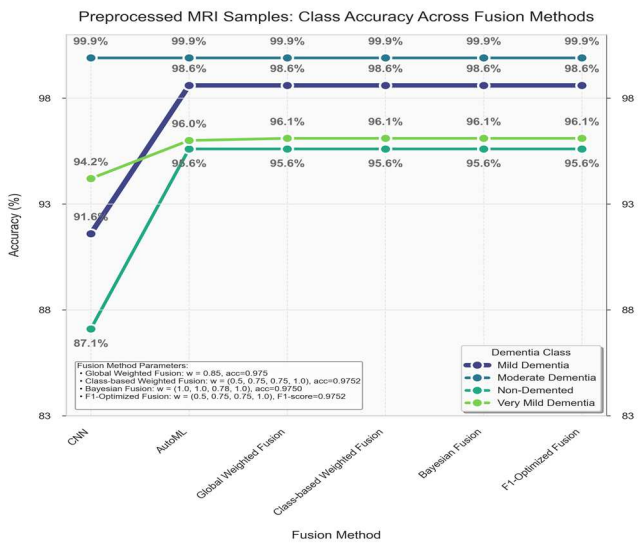

Fig. 4. Accuracy progress from the CNN stage through all the fusion stages for the augmented MRI branch [8]

On preprocessed skull-stripped sections, as seen in Fig. 4, fusion largely preserved the already strong performance of AutoML models, with further marginal improvements. Very Mild Dementia reached 96.0–96.1% accuracy, while Mild and Moderate Dementia consistently achieved 97.6–99.9%. Non-Demented remained stable at approximately 95.6%.

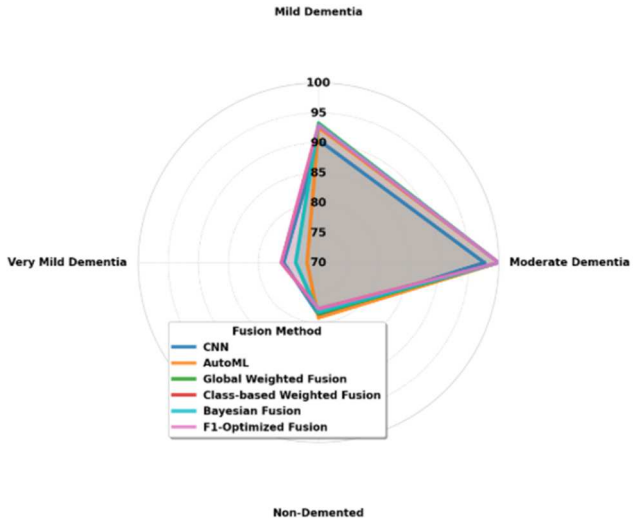

Fig. 5. Fusion methods performance [8]

Among the four tested strategies, class-based weighting and F1-optimized fusion consistently delivered the best balance across categories, suggesting that per-class tuning is more effective in the presence of class imbalance and variable difficulty, as seen also in Fig. 5, but the F1-score still remains the best overall in terms of performance.

B. Late Fusion – Bayesian Optimization – deep dive

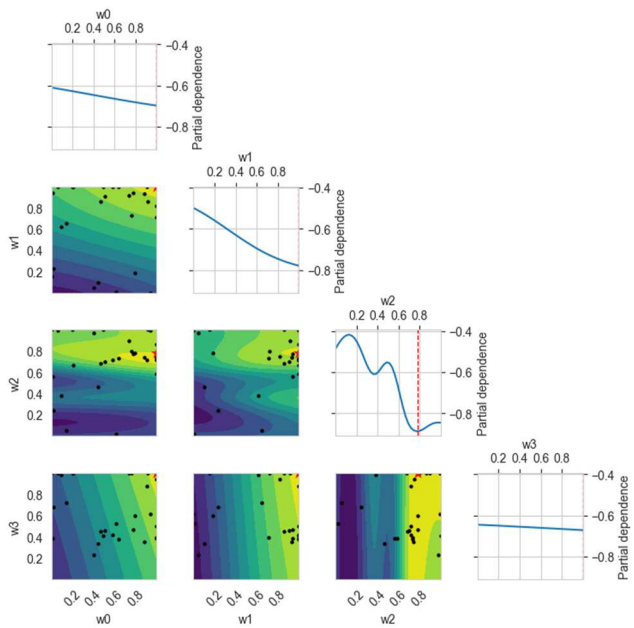

Fig. 6. Objective landscape [8]

A fusion method which is worth investigating more about is Bayesian optimization due to its adaptive weight search and consistently strong results. As presented in Fig. 6, the objective landscape of the optimization process illustrates how class-specific weights (w0–w3 corresponding to the classes from Fig. 4 in order) influenced overall performance [8]. The plots highlight smooth, monotonic effects for Mild

Authorized licensed use limited to: Tsinghua University. Downloaded on January 12,2026 at 07:01:38 UTC from IEEE Xplore. Restrictions apply.

and Moderate Dementia (w0, w1), contrasted with a non-linear response for Non-Demented (w2), where an optimum emerged near 0.78.

C. MMSE Regression

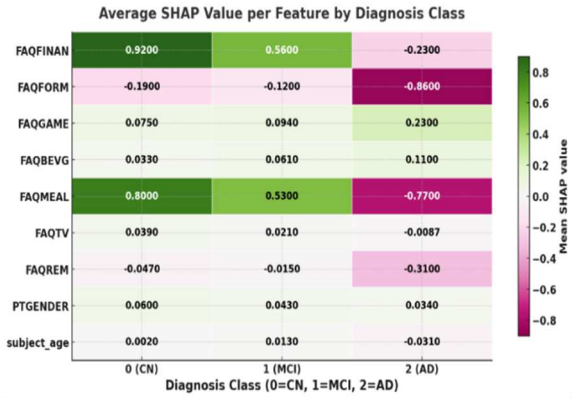

Fig. 7. MMSE features weight among classes [8]

The MMSE regression section aims to estimate patients' cognitive performance from clinical and demographic variables. This component complements the imaging classifiers by modeling functional aspects of cognition that are not directly observable in MRI data, using for this task FAQ-based activity scores and demographic features.

To better understand which variables contributed the most to the regression outcome, Fig. 7 shows the average feature importance by diagnostic class after mapping the original four labels (Non-Demented, Very Mild, Mild, Moderate) into the three ADNI categories (CN, MCI, AD). The analysis highlights the strong negative influence of FAQFINAN and FAQMEAL for patients in the AD class, while these same features display positive values in CN, confirming their role as early signs of cognitive decline. Other variables, such as FAQFORM and FAQREM, also gain importance as the disease progresses, aligning with expected cognitive deterioration patterns.

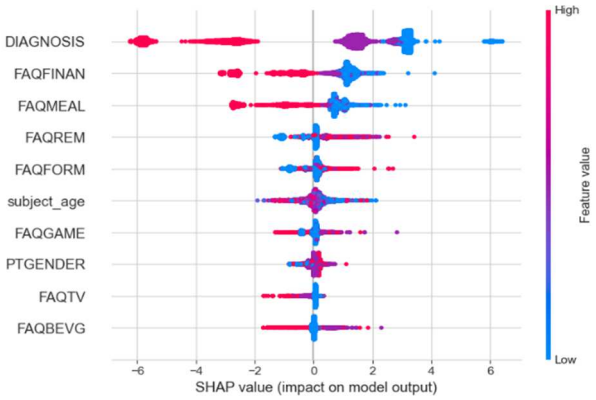

Fig. 8. MMSE global feature distribution [8]

The global distribution shown in Fig. 8 provides complementary evidence on overall feature impact across all samples. Diagnosis, derived from the imaging classifier, emerged as the most influential predictor of MMSE, followed closely by FAQFINAN and FAQMEAL, confirming the integration of structural and clinical signals within the model. Demographic factors such as age and gender contributed minimally, underscoring that clinical functionality and predicted diagnosis are more decisive in estimating cognitive performance.

D. Multimodal System

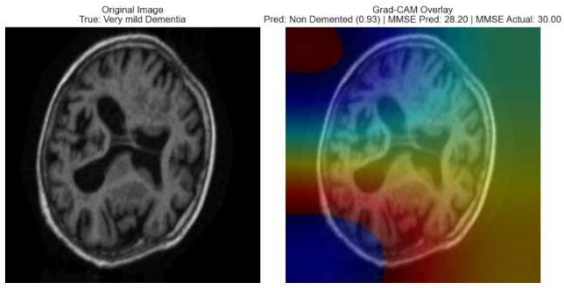

Fig. 9. Multimodal prediction [8]

The multimodal stage demonstrates the added value of integrating image-based classification outputs with independently trained MMSE regression predictions. Fig. 9 illustrates a representative case where the CNN alone misclassified a patient as Non-Demented with high confidence (0.93), despite the true label being Very Mild Dementia. However, the regression model estimated an MMSE score of 28.20, slightly below the maximum of 30, indicating subtle cognitive decline. This discrepancy highlights how structural features visible to the CNN may not capture early deterioration, while the MMSE score does.

The Grad-CAM overlay further shows that the CNN concentrated on broader cortical regions without fully detecting the subtle abnormalities linked to early dementia. In contrast, the regression output acted as a corrective layer, flagging a decline consistent with the clinical label, thus reducing the risk of overconfidence.

E. Literature Comparison

In this section, the proposed system is evaluated against other approaches from the literature, with the analysis divided into two parts: image classification and MMSE regression. The comparison considers both per-class accuracy for different dementia stages and overall multiclass performance, as well as the ability to predict MMSE scores based on multimodal features.

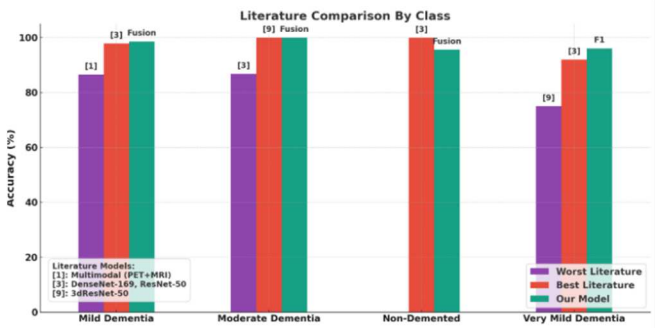

Fig. 10. Literature comparison by class [8]

From the perspective of per-class classification, as presented in Fig. 10, the fusion-based approach achieved strong results across all diagnostic categories. For Mild Dementia, accuracy reached 98.6%, slightly higher than the best literature value of 97.86% reported using DenseNet-169 and ResNet-50 [3]. For Moderate Dementia, accuracy achieved a perfect 100%, clearly exceeding multimodal PET+MRI CNN models, which reached only 86.51% [1]. Non-Demented patients were recognized at 95.6%, competitive with the 99.93% benchmark [3]. For Very Mild Dementia, the system obtained 96.1%, outperforming [3]

Authorized licensed use limited to: Tsinghua University. Downloaded on January 12,2026 at 07:01:38 UTC from IEEE Xplore. Restrictions apply.

(91.97%) and [9] (75.0%), demonstrating the advantage of optimized late fusion for subtle stages.

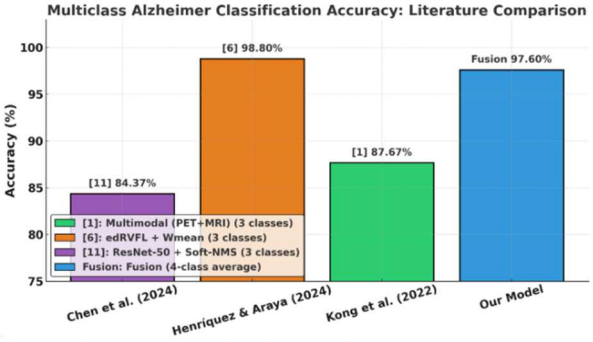

Fig. 11. Literature comparison by overall accuracy [8]

When comparing overall multiclass performance, as seen in Fig. 11, the proposed framework achieved 97.6% average accuracy across four categories. This result is close to the ensemble RVFL method of Henríquez & Araya [6], which reported 98.8% but in a simpler three-class setting (AD/CN/MCI). Other studies, such as [11] with ResNet-50 + Soft-NMS and [1] with PET-based CNNs, reached only 84.37% and 87.67%, respectively. The proposed system thus demonstrates robustness in handling four classes, including early dementia, which many prior works exclude.

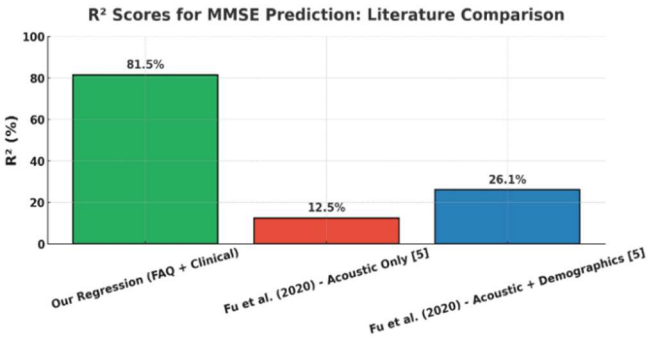

Fig. 12. MMSE score literature comparison [8]

In terms of MMSE regression, the Gradient Boosted Trees model achieved 81.5%, in terms of R^2, as seen in Fig. 12. This performance far exceeded [5], where acoustic-only features obtained just 12.5% of MMSE variance, and even the extended acoustic and demographic approach improved only to 26.1%. The clear superiority of the proposed regression approach underscores the benefit of combining clinical data (FAQ scores) with imaging-derived diagnosis.

V. Conclusion and Future Directions

This paper introduces a multimodal deep learning framework for Alzheimer's disease diagnosis and MMSE prediction, combining MRI imaging with clinical features. The dual CNN branches, refined through AutoML and late fusion, achieved 97.6% accuracy across four dementia stages and an R^2 of 0.812 for MMSE regression. Explainability analyses using Grad-CAM and SHAP confirmed that both structural and functional markers contributed meaningfully to predictions, enhancing clinical interpretability. Comparisons with state-of-the-art methods demonstrated superior or competitive performance, particularly for challenging categories such as Very Mild Dementia.

Limitations remain, notably the restricted availability of volumetric MRI data, which limited the effectiveness of 3D CNNs and graph-based models. Future work will address these challenges by expanding multimodal datasets and incorporating additional imaging techniques such as PET, DTI, and fMRI, alongside more advanced graph-based architectures. Such extensions aim to further improve robustness and personalization, paving the way toward clinically deployable systems for neurodegenerative disease monitoring.

References

[1] Z. Kong, M. Zhang, W. Zhu, Y. Yi, T. Wang, and B. Zhang, "Multi-modal data Alzheimer's disease detection based on 3D convolution," Biomed Signal Process Control, vol. 75, May 2022, doi: 10.1016/j.bspc.2022.103565.

[2] A. A. Tahami Monfared, M. J. Byrnes, L. A. White, and Q. Zhang, "Alzheimer's Disease: Epidemiology and Clinical Progression," Jun. 01, 2022, Adis. doi: 10.1007/s40120-022-00338-8.

[3] W. Al Shehri, "Alzheimer's disease diagnosis and classification using deep learning techniques," PeerJ Comput Sci, vol. 8, 2022, doi: 10.7717/PEERJ-CS.1177.

[4] Y. M. Choe, B. C. Lee, I. G. Choi, G. H. Suh, D. Y. Lee, and J. W. Kim, "Mmse subscale scores as useful predictors of ad conversion in mild cognitive impairment," Neuropsychiatr Dis Treat, vol. 16, pp. 1767–1775, 2020, doi: 10.2147/NDT.S263702.

[5] Z. Fu, F. Haider, and S. Luz, "Predicting Mini-Mental Status Examination Scores through Paralinguistic Acoustic Features of Spontaneous Speech," in Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS, Institute of Electrical and Electronics Engineers Inc., Jul. 2020, pp. 5548–5552. doi: 10.1109/EMBC44109.2020.9175379.

[6] P. A. Henríquez and N. Araya, "Multimodal Alzheimer's disease classification through ensemble deep random vector functional link neural network," PeerJ Comput Sci, vol. 10, 2024, doi: 10.7717/PEERJ-CS.2590.

[7] I. J. Khan et al., "Enhanced ROI guided deep learning model for Alzheimer's detection using 3D MRI images," Inform Med Unlocked, vol. 56, Jan. 2025, doi: 10.1016/j.imu.2025.101650.

[8] C.-I. Casapu, "Abordări inovative privind clasificarea bolilor degenerative cerebrale", Master Thesis, University "Dunarea de Jos", Galati, Romania, 2025

[9] S. Goel, N. Sharma, and Z. Zheng, "Diagnosis of Alzheimer's Disease Using 3D and 2D Convolutional Neural Networks," 2023.

[10] C.-I. Casapu and S. Moldovanu, "Classification of Microorganism Using Convolutional Neural Network and H2O AutoML," SYSTEM THEORY, CONTROL AND COMPUTING JOURNAL, vol. 4, no. 1, pp. 15–21, Mar. 2024, doi: 10.52846/stccj.2024.4.1.60.

[11] Y. Chen et al., "Automated Alzheimer's disease classification using deep learning models with Soft-NMS and improved ResNet50 integration," J Radiat Res Appl Sci, vol. 17, no. 1, p. 100782, Mar. 2024, doi: 10.1016/j.jrras.2023.100782

Authorized licensed use limited to: Tsinghua University. Downloaded on January 12,2026 at 07:01:38 UTC from IEEE Xplore. Restrictions apply.